

Project Demonstration

Date	4 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Project Demonstration Overview

This section demonstrates the complete workflow of the **Voice-Based Concept Understanding Analyzer**. The application accepts a student's voice explanation, converts speech into text, performs semantic analysis, evaluates conceptual understanding, and generates a detailed PDF report.

1. System Architecture

DATA FLOW DIAGRAM (LEVEL 0)

Voice-Based Concept Understanding Analyzer

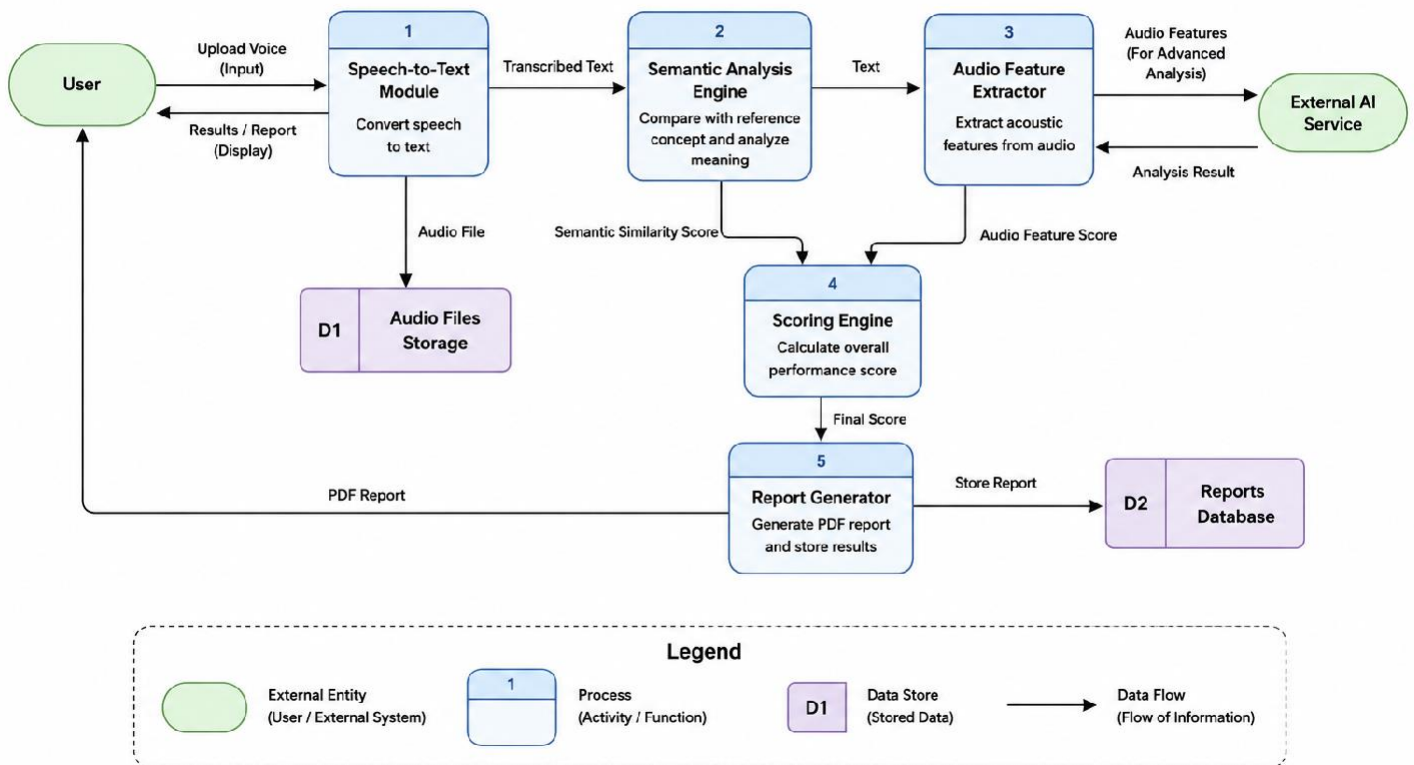


Figure 1: Data Flow Diagram

2. Project Structure

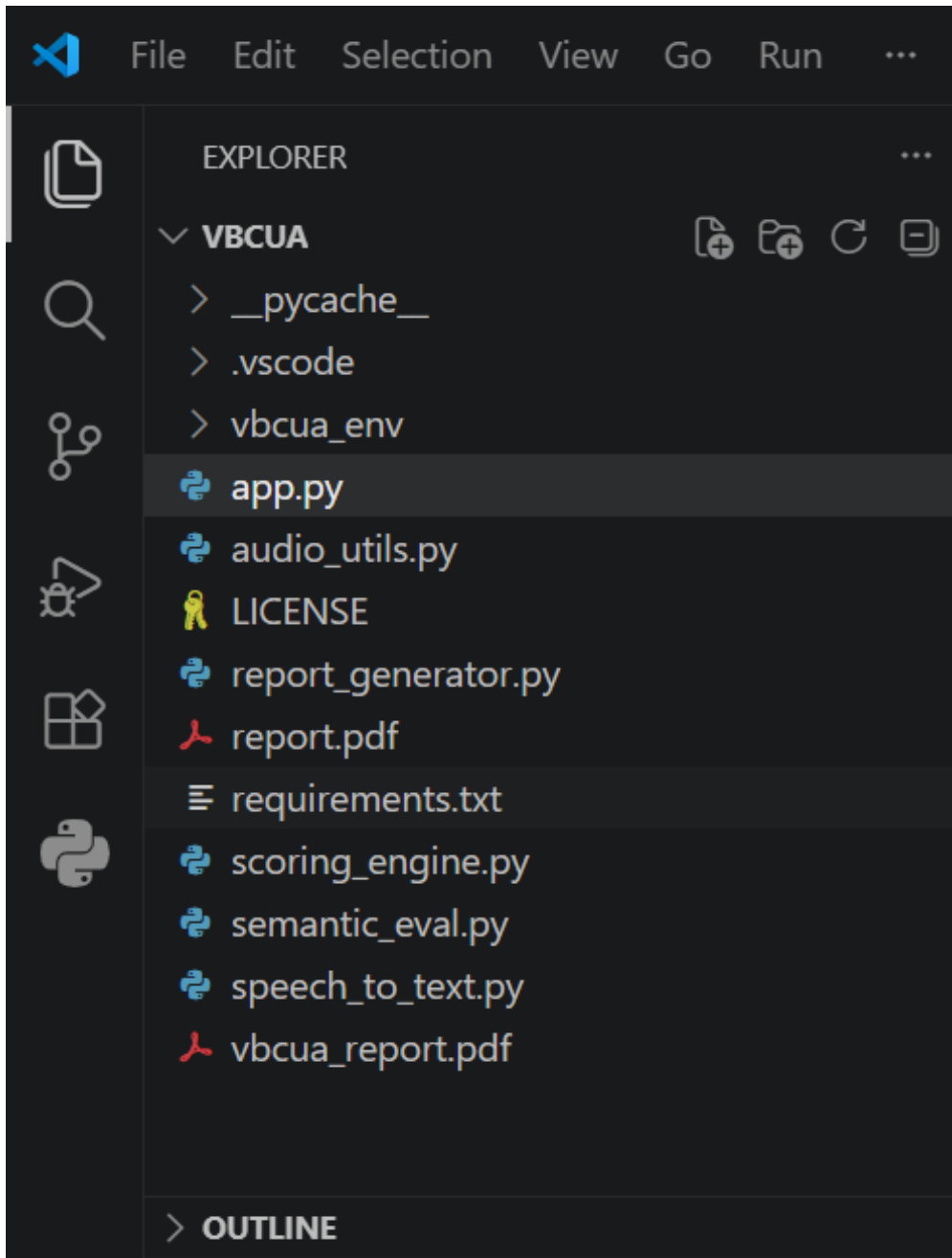


Figure 2: Project Folder Structure in VS Code

3. Running the Application

```
PS D:\Voice Based Concept Understanding Analysis\Source_Code> py -m streamlit run app.py
2026-07-05 09:38:17.995 Uvicorn server started on 0.0.0.0:8501
```

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>

Network URL: <http://192.168.43.185:8501>

Figure 3: Streamlit Server Started Successfully

4. Streamlit Application Running

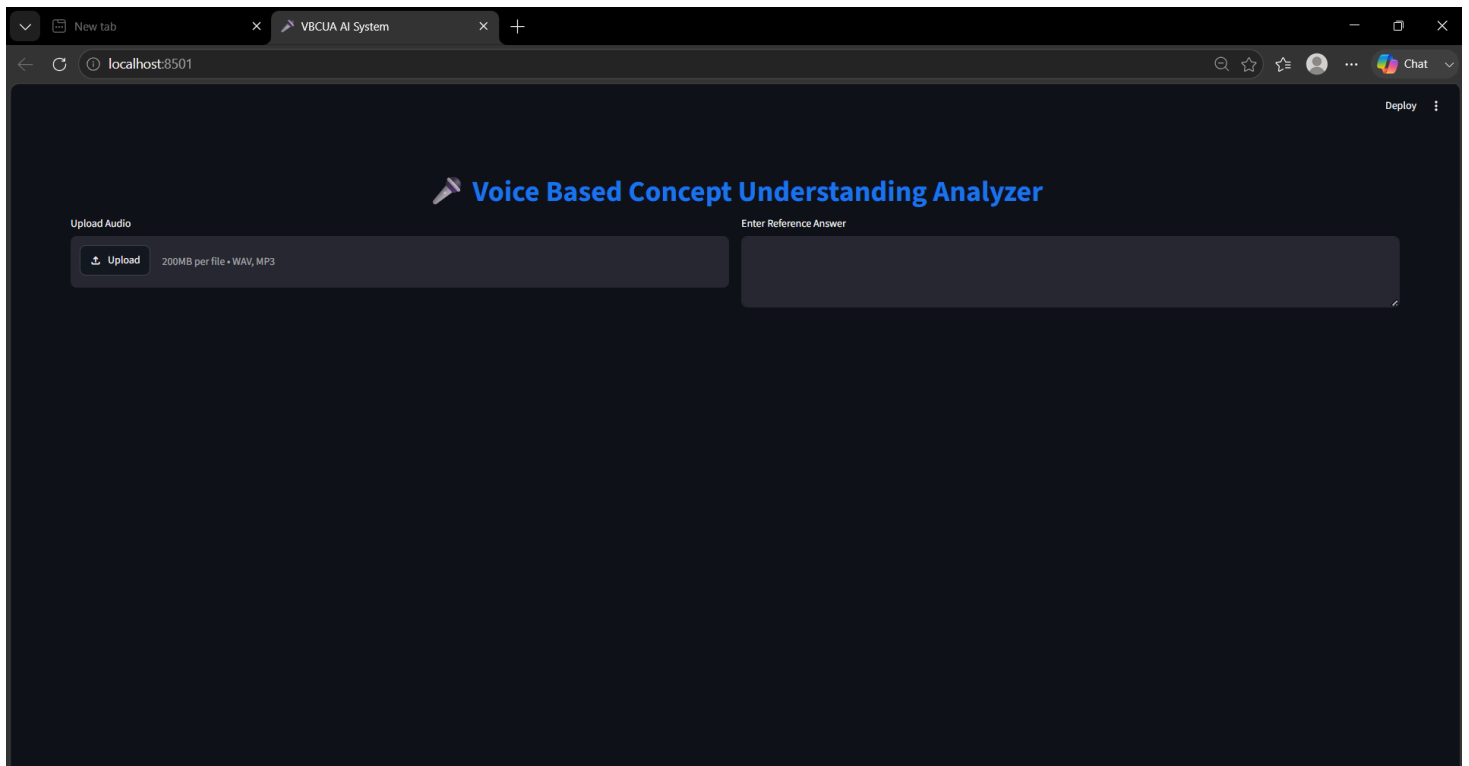


Figure 4 : Application Running Locahost

1. Application Workflow

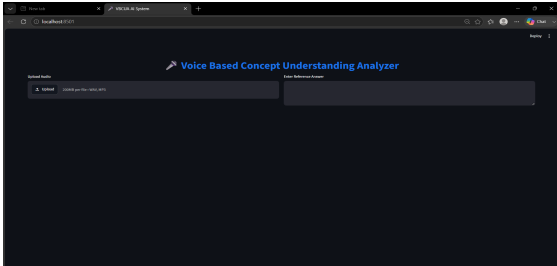


Figure 5: Home Page

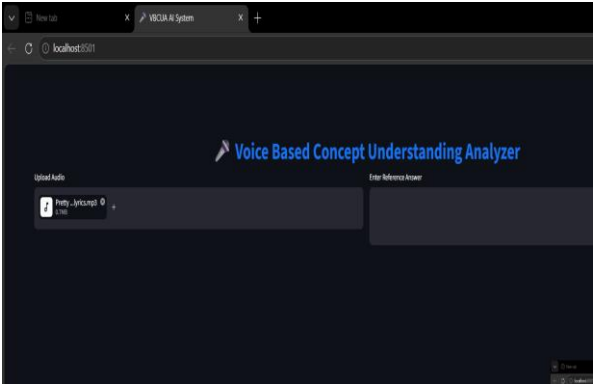


Figure 6: Audio Upload

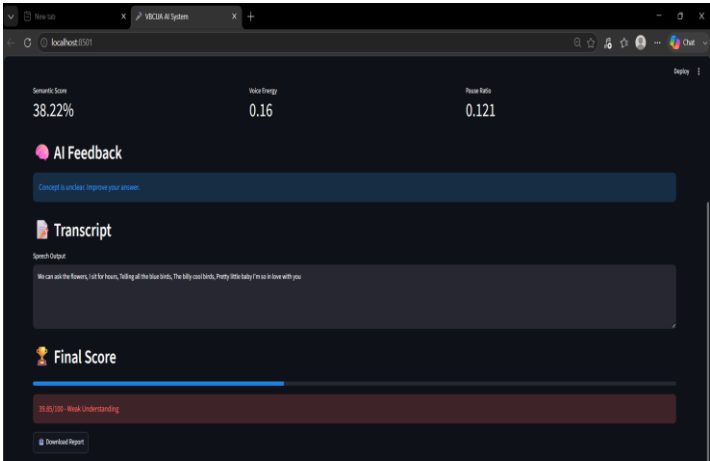


Figure 7: AI Evaluation Result

VBCUA Report

2026-07-10 18:34:55.961746

Result: 38.22% Understanding

Transcript:

We can ask the flowers, I sit for hours, Telling all the blue birds, The billy cool birds, Pretty little baby I'm so in love with you

Figure 9: Generated PDF Report

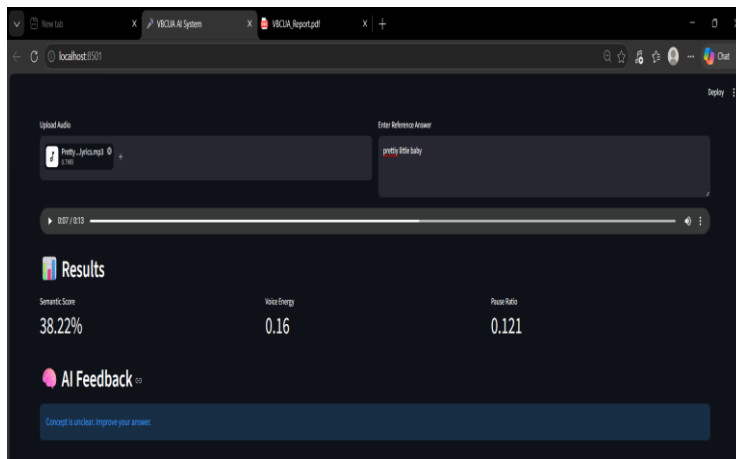


Figure 10: Strength & Semantic Analysis

2. Demonstration Summary

The **Voice-Based Concept Understanding Analyzer (VBCUA)** is an AI-powered web application developed to evaluate a user's conceptual understanding through spoken responses. During the demonstration, the user accessed the application using the Streamlit web interface and uploaded or recorded an audio response for a given question.

The system first converted the speech into text using **OpenAI Whisper Speech-to-Text** technology. The generated transcript was then compared with the expected answer using **Sentence Transformers** to measure semantic similarity and evaluate conceptual understanding rather than simple keyword matching.

The application also extracted important audio features such as speech clarity, speaking rate, and other voice characteristics using **Librosa**. Based on both semantic analysis and audio analysis, the system calculated an overall performance score.

